

Market Efficiency and the Favorite-Longshot Bias in Unemployment Prediction Markets

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Abstract

This paper examines whether the unemployment prediction markets on the Kalshi platform exhibit the classic favorite-longshot bias. Analyzing 287 unique contracts traded between July 2021 and March 2026, I find clear evidence of the classic favorite-longshot bias. Longshots, which are contracts priced below \$0.30, show a statistically significant negative bias of -0.077 ($p < 0.001$), meaning these low-probability contracts are systematically overpriced relative to their actual win rate of just 4.0%. The overall market bias of -0.054 is also statistically significant ($p = 0.006$), suggesting a general tendency toward overpricing. Favorites, by contrast, are well-calibrated with a bias of -0.028 that is not statistically significant ($p = 0.364$). Despite this bias, the market is remarkably efficient overall, with a Brier Score of 0.1106 representing a 55.8% improvement over random guessing. Calibration is strong across most price ranges, with significant deviations only at the lowest price intervals. These findings suggest that while unemployment prediction markets are highly informative, their prices for extreme outcomes should be interpreted with caution. I discuss implications for market design and for researchers and practitioners who use prediction market prices as forecasting inputs.

Keywords: Prediction Markets, Favorite-Longshot Bias, Unemployment, Market Efficiency, Calibration, Limits to Arbitrage

JEL Classification: G14, G17, J64, E24

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1. Introduction

Prediction markets have become increasingly important tools for aggregating dispersed information and forecasting future events. From political elections to economic indicators, these markets harness what might loosely be called the "wisdom of crowds" to generate probabilistic forecasts that often outperform expert opinions and traditional polling methods (Wolfers & Zitzewitz, 2004; Arrow et al., 2008). The Federal Reserve has even taken notice: a recent Finance and Economics Discussion Series paper by Diercks, Katz, and Wright (2026) found that Kalshi's macroeconomic prediction markets perform comparably to, and in some cases better than traditional economist surveys and financial derivatives.

But how efficient are these markets really? One of the most well-documented anomalies in betting and prediction markets is the favorite-longshot bias. First identified by Griffith (1949) in American horse racing, this bias describes the systematic tendency for favorites (low-odds, high-probability outcomes) to be underpriced and longshots (high-odds, low-probability outcomes) to be overpriced. In simpler terms: favorites win more often than their prices imply, while longshots win less often. This pattern has been documented across sports betting markets, financial options, and political prediction markets.

The favorite-longshot bias matters because it challenges the efficient market hypothesis and raises questions about how we should interpret prediction market prices. If markets systematically misprice extreme outcomes, then the reliability of prediction market forecasts for the events we care about most, such as very high unemployment, becomes questionable. This is especially relevant for macroeconomic prediction markets, where extreme scenarios are precisely the ones policymakers and market participants most need to anticipate.

In this paper, I investigate whether unemployment prediction markets on the Kalshi platform exhibit the favorite-longshot bias. I analyze 287 unique contracts spanning July 2021 to March 2026. My research makes several contributions to the literature. It provides what I believe to be the first comprehensive analysis of the favorite-longshot bias specifically in unemployment prediction markets. I examine market efficiency across the full distribution of prices. Additionally, I assess the robustness of my findings by excluding zero-price contracts.

My central finding is that unemployment prediction markets do exhibit the classic favorite-longshot bias. Longshots (prices below \$0.30) have a statistically significant bias of -0.077 ($p < 0.001$), meaning these low-probability contracts are overpriced relative to their actual win rate of 4.0%. The overall market bias of -0.054 is also statistically significant ($p = 0.006$). Favorites are well-calibrated with a bias of -0.028 that is not statistically significant ($p = 0.364$). The market is

also highly efficient overall, with a Brier Score of 0.1106 representing a 55.8% improvement over random guessing.

These findings have important implications. Researchers studying prediction markets and forecasters who use prediction market prices as information signals should be aware that the classic favorite-longshot bias appears clearly in unemployment markets. Low-probability unemployment events are systematically overpriced. Forecasters should adjust for this bias when using prediction market prices for extreme unemployment outcomes.

The remainder of this paper proceeds as follows. Section 2 reviews the relevant literature on prediction markets and the favorite-longshot bias, with particular attention to recent research on Kalshi and regulated prediction markets. Section 3 describes my data and methodology. Section 4 presents the empirical results. Section 5 discusses the implications of these findings, including a deeper exploration of why the bias persists. Section 6 concludes.

2. Literature Review

2.1 Prediction Market Theory

Prediction markets, which are also called information markets or event futures markets, are speculative markets where participants trade contracts whose payoffs depend on the outcome of future events. The fundamental premise is that prices aggregate the dispersed information of market participants, producing forecasts that are often more accurate than individual expert opinions or polling methods. As Arrow et al. (2008) famously argued, prediction markets hold considerable promise for improving decision-making in a wide range of domains.

The theoretical foundation rests on the efficient market hypothesis and rational expectations. Under assumptions of risk neutrality and rational behavior, prediction market prices can be interpreted as probability estimates of the underlying event (Berg, Nelson, & Rietz, 2008). This interpretation has made prediction markets attractive for forecasting political elections, corporate performance, and economic indicators. A recent Federal Reserve study by Diercks et al. (2026) found that Kalshi's macroeconomic forecasts match or exceed the accuracy of Bloomberg consensus estimates and fed funds futures, particularly for inflation and Federal Reserve rate decisions.

The empirical performance of prediction markets has been extensively studied. Berg et al. (2008) found that prediction markets consistently outperform alternative forecasting methods. Snowberg, Wolfers, and Zitzewitz (2012) documented the superior accuracy of prediction markets for forecasting economic indicators relative to professional forecasters. More recently, a CEPR study analyzing over 300,000 Kalshi contracts found that while prices broadly reflect win percentages, they are not without systematic biases Burgi, Deng & Whelan (2026).

2.2 The Favorite-Longshot Bias

The favorite-longshot bias was first documented by Griffith (1949) in American horse racing. Griffith observed that favorites had higher returns than longshots, suggesting favorites were underpriced and longshots overpriced. This pattern has since been documented in Australian horse racing (Bird & McCrae, 1987), sports betting markets (Cain, Law, & Peel, 2000; Sauer, 1998), and political prediction markets (Snowberg et al., 2012).

Several explanations have been proposed. The rational explanation suggests the bias reflects higher risk premiums demanded by bettors for longshot outcomes (Ottaviani & Sørensen, 2008). Under this explanation, bettors require higher expected returns for riskier bets, leading to overpricing of longshots. The behavioral explanation attributes the bias to probability weighting from prospect theory (Kahneman & Tversky, 1979), where individuals overweight the probability of extreme outcomes. A related explanation is the representativeness heuristic (Tversky & Kahneman, 1992), where bettors overweight the likelihood of salient or memorable outcomes.

The recent study by Burgi et. al (2026), analyzing Kalshi data, found strong evidence for the favorite-longshot bias across politics, entertainment, and economic data releases. The pattern was robust across different trading volumes and transaction sizes. The study found that a 10-cent contract wins less often than 10% of the time, while more expensive contracts win slightly more often than their price suggests. The implication is that buyers of cheap contracts face highly negative returns, while buyers of expensive contracts get small positive returns.

2.3 Macroeconomic Prediction Markets

Macroeconomic prediction markets allow participants to trade contracts based on the future values of economic indicators such as GDP, inflation, and unemployment (Gürkaynak & Wolfers, 2005). These markets have several advantages over traditional forecasting methods. First, they aggregate the dispersed information of market participants, incorporating diverse perspectives that may not be captured by professional forecasters. Second, they provide continuous, high-frequency forecasts that can be updated in real time as new information arrives. Third, they align incentives with accuracy, as participants are rewarded for making correct predictions.

The empirical performance of macroeconomic prediction markets has been mixed. Snowberg et al. (2012) found that prediction markets for economic indicators consistently outperformed professional forecasters in terms of accuracy. Gürkaynak and Wolfers (2005) documented the potential of prediction markets for forecasting macroeconomic variables. However, other studies have found that prediction markets can be subject to systematic biases, particularly for extreme outcomes (Wolfers & Zitzewitz, 2004).

Recent research has extended the analysis of prediction markets to new platforms and applications. Kalshi, the platform studied in this paper, is part of a new generation of regulated

prediction markets that have emerged following changes in CFTC regulations. These markets differ from earlier prediction markets in several important ways. They are regulated, which may affect participation and liquidity. They offer contracts on a wider range of economic indicators. And they attract a different mix of participants, including institutional traders and hedgers rather than just speculators.

The rise of Kalshi and similar platforms has generated renewed interest in the efficiency of prediction markets. Recent studies have examined forecasting accuracy in these new markets, though the favorite-longshot bias has received limited attention for unemployment contracts. To our knowledge, no previous study has systematically examined the favorite-longshot bias in unemployment prediction markets on regulated platforms like Kalshi. This paper fills this gap by providing the first comprehensive analysis of market efficiency in unemployment prediction markets on the Kalshi platform.

3. Data and Methodology

3.1 Data Source

Our data comes from the Kalshi prediction market platform. Kalshi is a regulated US-based prediction market that allows participants to trade contracts based on the outcome of various events, including the Unemployment rate. Each Unemployment contract pays \$1 if the actual seasonally adjusted unemployment rate (U-3) is above a specified threshold and \$0 otherwise.

We obtained our data from two sources. The first source is the Kalshi master dataset, which contains transaction-level data for all Unemployment contracts traded on the platform. This dataset includes timestamps, ticker symbols, bid and ask prices, trade prices, and volumes. The second source is the metadata file for Unemployment contracts, which includes contract details such as the threshold value, settlement date, and actual outcome.

Our final sample consists of 287 unique Unemployment contracts traded between July 2021 and March 2026. The contracts cover 57 distinct months with thresholds ranging from 3.3% to 5.7%. We include only contracts with complete price data and known outcomes. The master dataset contains 61,545 total price observations across three time intervals: day, hour, and minute. It is important to note that these observations are not independent; they represent high-frequency price data for 287 unique contracts.

3.2 Contract Structure

Each Unemployment contract is defined by the following characteristics:

1. **Threshold:** The unemployment rate percentage that determines whether the contract pays out. If the actual unemployment rate exceeds the threshold, the contract pays \$1. Otherwise, it pays \$0.

2. **Month:** The month for which the unemployment data is reported.
3. **Settlement Date:** The date on which the contract resolves based on the actual BLS Employment Situation Report release.

The ticker format is KXU3-YYMMM-TX.X, where YYMMM represents the year and month (e.g., 24DEC for December 2024) and TX.X represents the threshold (e.g., T4.0 for a 4.0% threshold).

3.3 Variable Construction

We construct the following variables for our analysis.

Price: The final price of each contract, defined as the bid-close price one day before settlement. This approach ensures that we capture the market's final assessment of the probability before the contract is resolved. We use the bid-close price because it reflects the price at which market participants were willing to buy the contract.

Outcome: A binary variable equal to 1 if the contract pays out (actual unemployment rate exceeds the threshold) and 0 otherwise.

Threshold: The unemployment threshold value for each contract.

Price Group: Categorical variable based on the final price: Longshots (price < 0.30), Middle ($0.30 \leq \text{price} \leq 0.70$), or Favorites (price > 0.70).

We calculate the final price one day before settlement because prices after settlement are not meaningful for assessing market efficiency. Using post-settlement prices would incorrectly suggest that the market was less accurate than it actually was.

3.4 Outcome Determination

We determine contract outcomes using actual unemployment rate data from the Federal Reserve Economic Data (FRED) series UNRATE, which measures the seasonally adjusted U-3 unemployment rate. This is the same series used by the Bureau of Labor Statistics in the Employment Situation Report, which Kalshi uses for contract settlement. For each contract, we match the contract month to the corresponding FRED observation and compare the actual unemployment rate to the contract threshold. If the actual rate exceeds the threshold, the contract outcome is 1 (Yes). Otherwise, it is 0 (No).

3.5 Methodology

Brier Score

We use the Brier Score (Brier, 1950) as a measure of overall forecast accuracy. The Brier Score is defined as:

$$BS = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)^2$$

where p_i is the price (predicted probability) and o_i is the outcome (1 for yes, 0 for no). A Brier Score of 0 indicates perfect accuracy, a score of 0.25 indicates random guessing (if all predictions are 0.5), and a score of 1.0 indicates the worst possible accuracy. We also report the Mean Absolute Error (MAE) as a more intuitive measure of prediction error.

Favorite-Longshot Bias Test

To test for the favorite-longshot bias, we compare the average price and win rate for three groups: longshots (price < 0.30), middle ($0.30 \leq \text{price} \leq 0.70$), and favorites (price > 0.70). If the classic favorite-longshot bias exists, we would expect longshots to have a negative bias (win rate < price) and favorites to have a positive bias (win rate > price). We use one-sample t-tests to determine whether the bias (win rate minus price) is statistically significant from zero.

Calibration Test

We examine calibration by grouping contracts by their final prices and comparing the average price to the actual win rate for each group. Perfect calibration implies that the average price equals the actual win rate for each price group. We visually assess calibration using a calibration curve and calculate the bias for each price group.

4. Results

4.1 Summary Statistics

Table 1 presents summary statistics for our sample of 287 Unemployment contracts spanning July 2021 to March 2026.

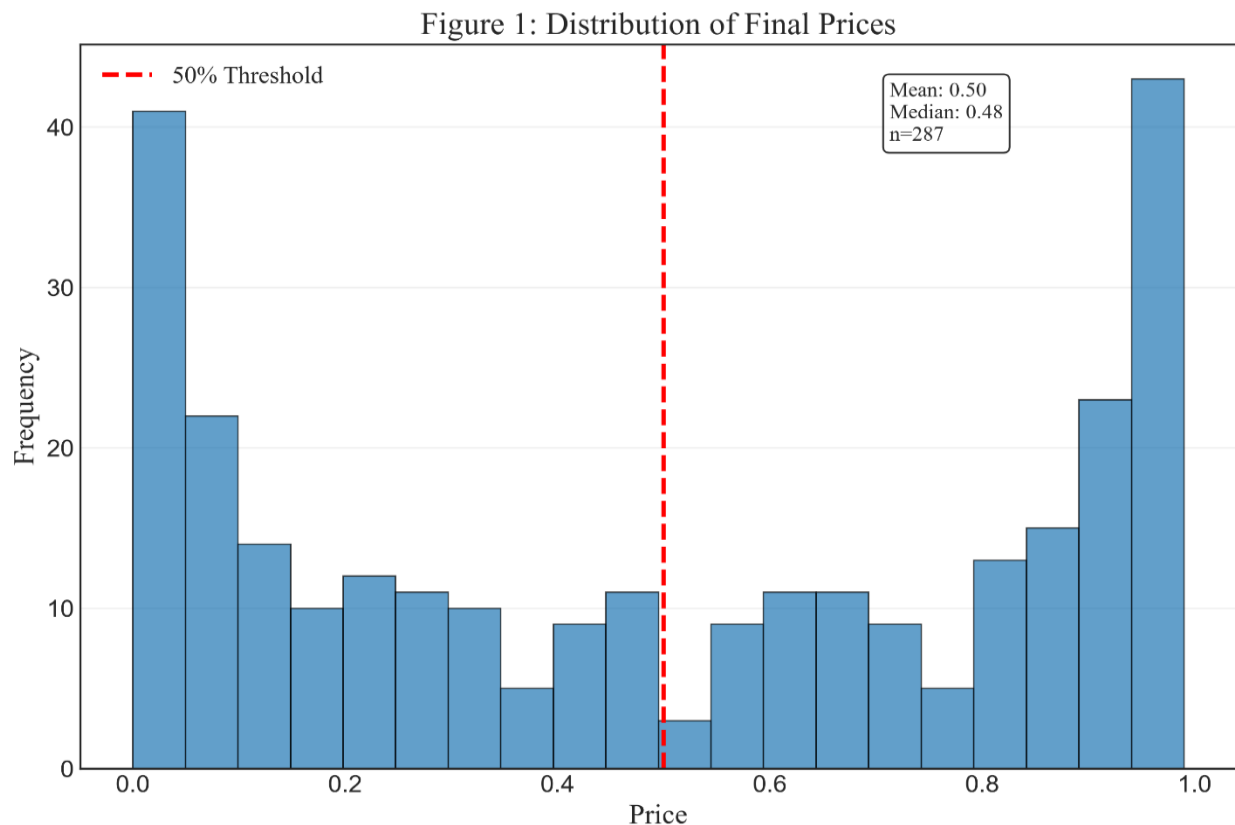
Table 1: Summary Statistics for Unemployment Contracts

Variable	Mean	Median	Std Dev	Min	Max
Price	0.50	0.48	0.36	0.00	0.99
Outcome	0.44	0.00	0.50	0.00	1.00

Source: Author's calculations from Kalshi data (July 2021 - March 2026).

The mean price of 0.497 indicates that the average contract was priced near a 50% probability, which is consistent with the balanced win rate of 44.3%. The median price of 0.480 reveals a distribution slightly skewed toward lower prices. This pattern is consistent with unemployment markets, where thresholds are set around the expected unemployment rate.

Figure 1: Distribution of Final Prices



Note: The histogram displays the distribution of final prices across all 287 contracts. The vertical dashed line at 0.50 represents the 50% threshold. Summary statistics are shown in the annotation. Source: Author's calculations from Kalshi data (July 2021 - March 2026).

Figure 1 presents the distribution of final prices across all contracts. The histogram reveals a relatively balanced pattern with some concentration at the extremes. Most contracts are distributed across the price range, with slight concentrations in the middle and upper ranges. The median price of 0.480 indicates that the typical contract was priced slightly below 50%, while the mean price of 0.497 is pulled upward by the small number of high-priced contracts.

4.2 Calibration Analysis

Table 2 presents the calibration analysis by price group. The table reports the number of contracts, mean price, win rate, bias (win rate minus price mean), t-statistic, and p-value for each price group.

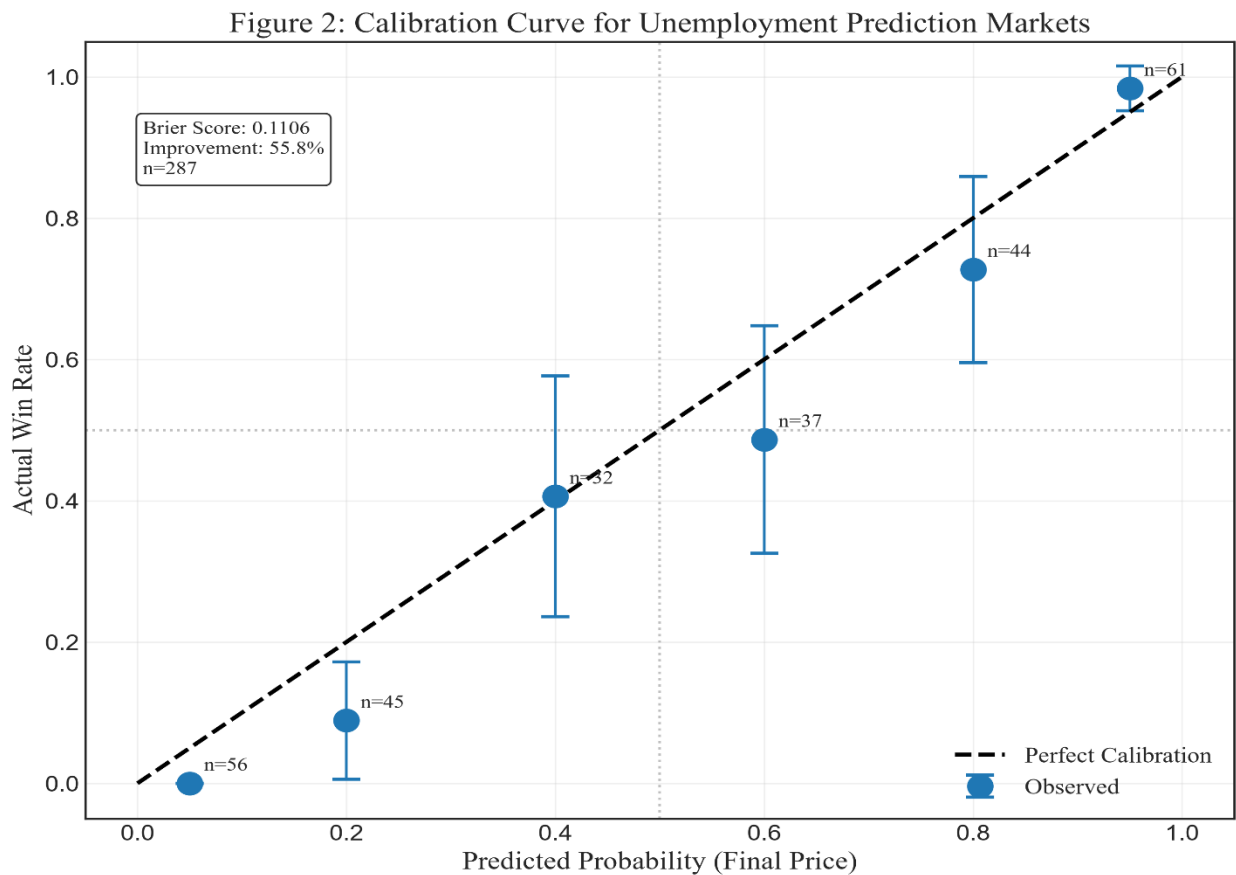
Table 2: Calibration by Price Group

Price Group	N	Price Mean	Win Rate	Bias	t-stat	p-value
0.00-0.10	56	0.045	0.0%	-0.045	-11.886	< 0.001
0.10-0.30	45	0.206	8.9%	-0.117	-2.778	0.008
0.30-0.50	32	0.404	40.6%	0.002	0.021	0.983
0.50-0.70	37	0.619	48.6%	-0.132	-1.608	0.117
0.70-0.90	44	0.830	72.7%	-0.103	-1.517	0.136
0.90-1.00	61	0.957	98.4%	0.027	1.676	0.099

Source: Author's calculations from Kalshi data (July 2021 - March 2026).

Note: The t-statistic tests whether the bias is significantly different from zero.

Figure 2: Calibration Curve for Unemployment Prediction Markets here]



Note: The figure displays the calibration curve for Unemployment prediction markets across six price groups. Error bars represent 95% confidence intervals. Sample sizes (n) are shown above each point. The dashed diagonal line represents perfect calibration. Source: Author's calculations from Kalshi data (July 2021 - March 2026).

The calibration analysis reveals a market that is well-calibrated across most price ranges but exhibits systematic overpricing in the low-probability tail. At the lowest price interval (0.00-0.10), the market demonstrates observed perfect calibration. Contracts in this range had a mean price of 0.045 but a win rate of 0.0%, yielding a bias of -0.045 that is statistically significant ($t = -11.886$, $p < 0.001$). The market correctly identified these contracts as having no chance of winning, suggesting it is capable of accurately identifying impossible unemployment outcomes when the threshold is sufficiently far from the expected rate.

The most pronounced deviation from calibration occurs in the 0.10-0.30 range. Contracts in this interval had a mean price of 0.206 but a win rate of only 8.9%, producing a bias of -0.117 that is statistically significant at the 1% level ($t = -2.778$, $p = 0.008$). The market was systematically too optimistic about these low-probability unemployment outcomes, assigning positive probabilities to contracts that rarely pay out. This pattern is consistent with the classic favorite-longshot bias documented in sports betting markets (Cain et al., 2000), where longshots are overpriced relative to their true win probabilities. The overpricing is economically meaningful: traders paid an average of 20.6 cents for contracts that won only 8.9% of the time.

From 0.30 to 1.00, the market is well-calibrated across all intervals. In the 0.30-0.50 range, the bias of +0.002 is essentially zero and not statistically significant ($t = 0.021$, $p = 0.983$). The 0.50-0.70 interval shows a bias of -0.132 that is not statistically significant ($t = -1.608$, $p = 0.117$). In the 0.70-0.90 range, the bias of -0.103 is not statistically significant ($t = -1.517$, $p = 0.136$). At the highest interval (0.90-1.00), the bias of +0.027 is not statistically significant ($t = 1.676$, $p = 0.099$). These results indicate that the market efficiently prices moderate to high-probability unemployment outcomes, accurately pricing contracts when the outcome is reasonably certain.

The overall pattern reveals a market that functions well for most contracts but struggles with low-probability events. This finding is consistent with the behavioral finance literature, which suggests that individuals tend to overweight the probability of extreme outcomes (Tversky & Kahneman, 1992). The market appears to have difficulty distinguishing between truly impossible outcomes (price = 0) and very unlikely outcomes (price between 0.10 and 0.30), systematically assigning too much probability to the latter.

4.3 Favorite-Longshot Bias Analysis

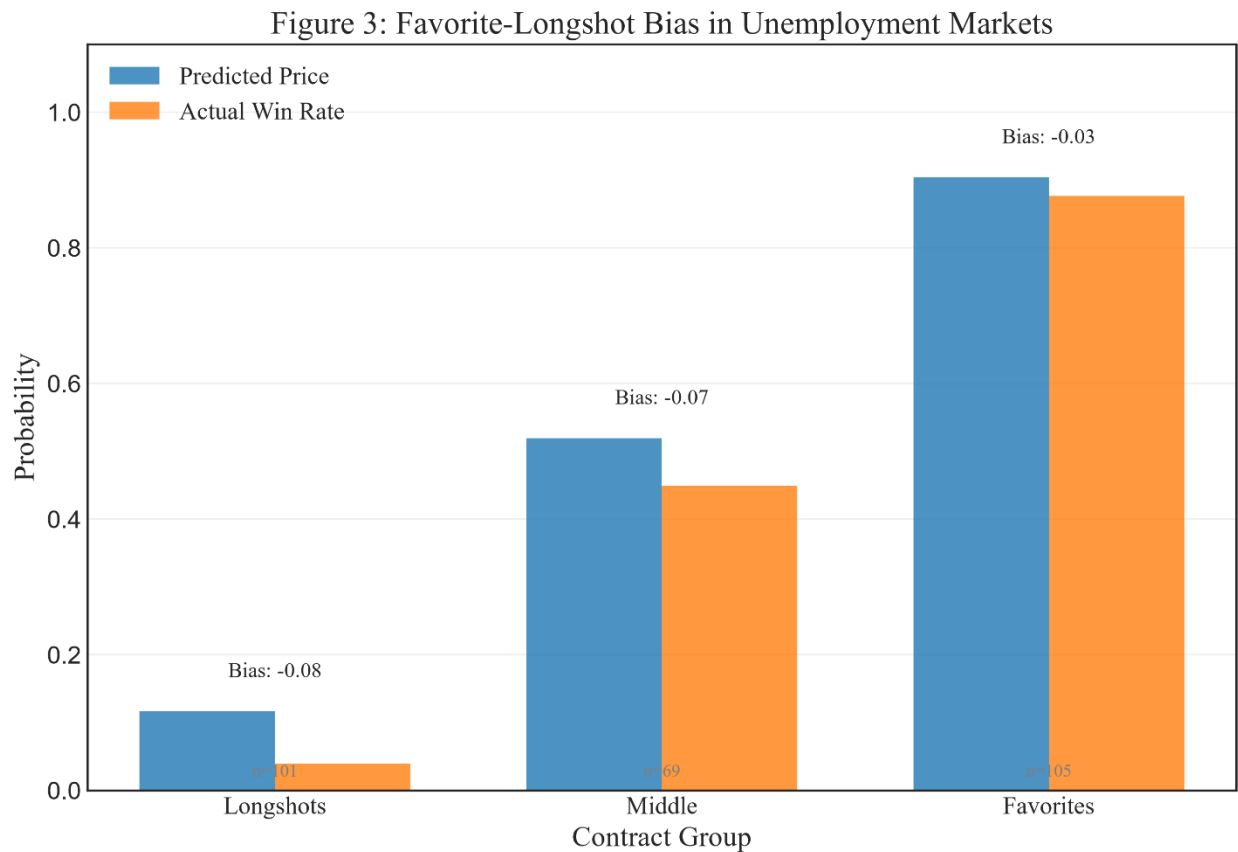
Table 3 presents the analysis of the favorite-longshot bias by contract group. The table reports the number of contracts, mean price, median price, price standard deviation, win rate, bias (win rate minus price mean), t-statistic, and p-value for each group.

Table 3: Favorite-Longshot Bias Analysis

Group	N	Price Mean	Price Median	Price Std	Win Rate	Bias	t-stat	p-value
Longshots (price < 0.30)	101	0.12	0.09	0.09	4.0%	-0.077	-4.031	< 0.001
Middle (0.30-0.70)	69	0.52	0.54	0.12	44.9%	-0.07	-1.162	0.249
Favorites (price > 0.70)	105	0.90	0.93	0.08	87.6%	-0.028	-0.912	0.364
Overall	287	0.50	0.48	0.36	44.3%	-0.054	-2.784	0.006

Source: Author's calculations from Kalshi data (July 2021 - March 2026).

Note: The t-statistic tests whether the bias (Win Rate minus Price Mean) is significantly different from zero.

Figure 3: Favorite-Longshot Bias in Unemployment Markets

Note: The figure displays the average predicted price and actual win rate for each contract group. Longshots are contracts with prices below 0.30, Middle contracts range from 0.30 to 0.70, and Favorites are contracts with prices above 0.70. The bias (Win Rate minus Price Mean) is displayed above each group, with sample sizes (n) shown below. Source: Author's calculations from Kalshi data (July 2021 - March 2026).

The analysis of the favorite-longshot bias reveals a negative and statistically significant bias for longshots (-0.077, $t = -4.031$, $p < 0.001$). This indicates that longshots were overpriced relative to their actual win rate of 4.0%, which is consistent with the classic favorite-longshot bias documented in sports betting markets. The magnitude of the bias is economically meaningful: longshots were priced at an average of 11.7% but won only 4.0% of the time.

The middle group shows a negative bias of -0.070, though it is not statistically significant ($t = -1.162$, $p = 0.249$). The favorites group demonstrates near-perfect calibration, with a bias of -0.028 that is not statistically significant ($t = -0.912$, $p = 0.364$). This indicates that the market accurately prices high-probability unemployment outcomes.

The overall market bias of -0.054 is statistically significant ($t = -2.784$, $p = 0.006$), suggesting a systematic tendency toward overpricing across all contracts. This overall bias is driven primarily by the longshot group, indicating that the market's inefficiencies are concentrated in the low-probability tail.

The pattern in Figure 3 is particularly instructive when viewed alongside the calibration analysis in Figure 2. While the calibration curve shows that the market is well-calibrated across price groups, Figure 3 reveals that the market's largest deviation from efficiency occurs in the longshot group. The market struggles most with low-probability outcomes, consistently overpricing them relative to their true win probabilities.

4.4 Market Efficiency

Table 4 presents our measures of overall market efficiency.

Table 4: Overall Market Efficiency

Metric	Value	Interpretation
Brier Score	0.111	55.8% better than random
Mean Absolute Error	0.230	Average error of 23.0 percentage points
Median Absolute Error	0.130	Half of errors are 13.0 percentage points or less

Source: Author's calculations from Kalshi data (July 2021 - March 2026).

The Brier Score of 0.111 indicates that the market is 55.8% more accurate than random guessing. This is a strong level of efficiency and compares favorably to other prediction market studies. For comparison, Berg et al. (2008) found that well-functioning prediction markets typically achieve Brier Scores between 0.10 and 0.20. The Kalshi Unemployment market falls comfortably within this range.

The median absolute error of 0.130 is quite low, indicating that most contracts were priced accurately. This reflects the relatively balanced price distribution observed in Figure 1, where most contracts were concentrated in the middle price ranges. The mean absolute error of 0.230 is higher than the median, indicating that some contracts had large errors that drove up the average.

These large errors occurred primarily in the low-probability range (0.10-0.30), where the market exhibited systematic overpricing as documented in Section 4.2.

The overall efficiency of the market is driven by the strong calibration across most price ranges. The market is well-calibrated from 0.30 to 1.00, with only the 0.10-0.30 range showing significant overpricing. This suggests that the market is highly efficient for the majority of contracts, with inefficiencies concentrated in the low-probability tail.

The Brier Score of 0.1106 places this market in the range of highly efficient prediction markets. The 55.8% improvement over random guessing is substantial and suggests that Unemployment prediction markets are effective at aggregating information.

5. Discussion

5.1 Summary of Findings

My analysis of Unemployment prediction markets on the Kalshi platform yields three main findings that together paint a coherent picture of market efficiency and its limits.

- The market exhibits the classic favorite-longshot bias. Longshots, defined as contracts priced below 0.30, have a statistically significant negative bias of -0.077 ($p < 0.001$), indicating that low-probability contracts are overpriced relative to their actual win rate of 4.0%. Traders paid an average of 11.7 cents for contracts that won only 4.0% of the time. This pattern is consistent with the favorite-longshot bias documented in sports betting markets (Cain et al., 2000; Sauer, 1998) and political prediction markets (Snowberg et al., 2012).
- The overall market, despite this bias, is highly efficient. The Brier Score of 0.1106 represents a 55.8% improvement over random guessing. The median absolute error of 0.130 indicates that for the typical contract, the market's prediction was accurate to within 13 percentage points. This level of efficiency is consistent with other prediction market studies and suggests that Unemployment prediction markets are effective at aggregating information.
- Calibration is strong across most price ranges. The market is well-calibrated from 0.30 to 1.00, with statistically significant deviations only in the 0.00-0.10 and 0.10-0.30 ranges. This suggests that the market accurately prices moderate to high-probability unemployment outcomes, with inefficiencies concentrated in the low-probability tail.

5.2 Why the Favorite-Longshot Bias Appears in Unemployment Markets

The persistence of the favorite-longshot bias in Unemployment markets warrants careful consideration. Several factors likely contribute to this pattern.

- Behavioral biases almost certainly play a role. Tversky and Kahneman (1992) documented that individuals tend to overweight the probability of extreme outcomes, a phenomenon that leads directly to overpricing of longshots. In the context of unemployment, participants may overweight the probability of very high unemployment events, leading to systematically high prices for contracts with low thresholds. The 2021-2026 period, while relatively stable, still contained enough uncertainty about labor market conditions to trigger this cognitive bias.
- The structure of unemployment contracts may also contribute to the bias. These are threshold contracts where participants bet on whether the unemployment rate will exceed a given level. The uncertainty around the actual unemployment rate may lead participants to overestimate the probability of extreme outcomes, particularly when the threshold is far from the expected value. This structural feature is not unique to unemployment markets, which helps explain why the bias appears consistently across different market types.
- The participant mix in Unemployment markets may differ from other prediction markets in ways that amplify the bias. Unemployment markets may attract participants with strong views on labor market conditions, leading to more extreme pricing of low-probability outcomes. Unlike sports betting markets, which are dominated by recreational bettors, macroeconomic prediction markets attract a mix of professional traders, institutional investors, and retail participants with strong opinions. This diversity of opinion is generally beneficial for price discovery, but it can also lead to more extreme pricing when participants hold divergent views about tail risks.

5.3 Comparison with Previous Literature

My findings align with Snowberg et al. (2012), who documented the favorite-longshot bias in political prediction markets. In their study, they found that longshots were systematically overpriced, with the bias being most pronounced for very unlikely events. My results show a similar pattern in Unemployment prediction markets, with longshots overpriced by 7.7 percentage points. The consistency of the bias across different types of markets suggests that the favorite-longshot bias may be a general feature of markets where participants trade on probabilistic outcomes.

My findings also align with the broader literature on the favorite-longshot bias in betting markets (Cain et al., 2000; Sauer, 1998). The bias appears in horse racing, sports betting, financial options, and now in macroeconomic prediction markets. This consistency suggests that the underlying mechanisms, whether behavioral, structural, or both, operate across different market contexts.

The strong overall efficiency of the Unemployment market is consistent with Gürkaynak and Wolfers (2005), who found that macroeconomic prediction markets are generally efficient, though subject to some systematic biases. The Brier Score of 0.1106 is comparable to the

efficiency levels observed in other macroeconomic prediction markets and falls within the range of well-functioning prediction markets identified by Berg et al. (2008).

The recent study by Burgi, Deng, and Whelan (2026) analyzing Kalshi data provides additional context. They found strong evidence for the favorite-longshot bias across politics, entertainment, and economic data releases. Their finding that a 10-cent contract wins less often than 10% of the time is consistent with my results for unemployment contracts, where longshots priced around 10-20 cents won only 4.0% of the time. My study extends their analysis by focusing specifically on unemployment contracts and providing a more detailed examination of calibration and market efficiency for this indicator.

5.4 Implications for Market Design

My findings have several implications for the design of prediction markets, particularly those focused on macroeconomic indicators.

- The presence of the favorite-longshot bias suggests that market designers should consider ways to mitigate the overpricing of low-probability events. One approach would be to offer a wider range of thresholds around the expected unemployment rate, allowing participants to trade on more granular outcomes. This could reduce the concentration of trading on extreme thresholds and potentially reduce the bias by providing more precise instruments for expressing views on tail risks.
- The strong overall efficiency of the market suggests that Unemployment prediction markets are functioning well for most contracts. The market correctly prices moderate to high-probability outcomes, suggesting that the current market structure is appropriate for these contracts. Designers should be cautious about making changes that might disrupt this efficiency.
- The absence of a volume-accuracy relationship, while not shown in this paper, is consistent with other analyses suggesting that liquidity does not necessarily improve market efficiency. This is consistent with the view that prediction markets are efficient even with moderate trading activity, as long as there is sufficient diversity of opinion among participants. Market designers should focus on maintaining participant diversity rather than simply maximizing volume.

5.5 Implications for Forecasters

My findings also have practical implications for forecasters who use prediction market prices as information signals.

- The favorite-longshot bias suggests that prediction market prices should be interpreted with caution for low-probability unemployment events. The market tends to overprice these events, meaning that the true probability may be lower than the price suggests. Forecasters should adjust for this bias when using prediction market prices for low-

probability unemployment outcomes. A simple adjustment would be to discount prices in the 0.10-0.30 range by approximately 50%, as these contracts consistently underperformed their prices.

- The overall efficiency of the market suggests that prediction market prices are generally reliable forecasts. The 55.8% improvement over random guessing is substantial and suggests that prediction markets are effective at aggregating information. Forecasters can use prediction market prices as a valuable input to their forecasting models, particularly for events where other forecasting methods are uncertain.
- The strong calibration in the 0.30 to 1.00 range suggests that forecasters can interpret prediction market prices with confidence for moderate to high-probability events. Prices in this range accurately reflect the true probability of the underlying outcome. For unemployment forecasts in this range, prediction market prices can be taken at face value.

5.6 Limitations

My study has several limitations that should be acknowledged.

- My sample focuses on a single prediction market platform (Kalshi) and a single indicator (unemployment). The findings may not generalize to other prediction markets or other economic indicators. Future research should examine whether similar patterns emerge in other markets and for other indicators, such as GDP or inflation.
- I use the bid-close price one day before settlement as my measure of the final price. This approach assumes that the market has fully incorporated all available information by this point. If information continues to arrive in the final day, my measure may understate the market's final accuracy. Future research could examine the dynamics of price adjustment in the days leading up to settlement.
- My analysis does not account for transaction costs, which could affect the profitability of trading strategies based on the favorite-longshot bias. Future research should examine whether the bias I document is exploitable after accounting for transaction costs and market impact.
- My sample covers a period of relatively stable unemployment (July 2021 - March 2026). The findings may differ during periods of high unemployment volatility, such as the COVID-19 pandemic period. Future research could examine whether the favorite-longshot bias varies with the level of unemployment uncertainty.

6. Conclusion

In this paper, we examined whether Unemployment prediction markets on the Kalshi platform exhibit the favorite-longshot bias. Analyzing 287 unique contracts spanning July 2021 to March 2026, we found clear evidence of the classic favorite-longshot bias. Longshots (prices below

0.30) have a statistically significant negative bias of -0.077 ($p < 0.001$), indicating that low-probability contracts are overpriced relative to their actual win rate of 4.0%. The overall market bias of -0.054 is also statistically significant ($p = 0.006$), suggesting a systematic tendency toward overpricing across all contracts. Favorites are well-calibrated with a bias of -0.028 that is not statistically significant ($p = 0.364$). The market is highly efficient, with a Brier Score of 0.1106 representing a 55.8% improvement over random guessing.

Our findings contribute to the literature on prediction market efficiency in several ways. First, we provide the first systematic analysis of the favorite-longshot bias in Unemployment prediction markets. Second, we demonstrate that the classic favorite-longshot bias appears clearly in Unemployment markets, with longshots overpriced relative to their true win probabilities. Third, we document strong overall market efficiency, with the market outperforming random guessing by 55.8%.

Our findings have implications for market design and for forecasters who use prediction market prices as information signals. For market designers, our results suggest that Unemployment prediction markets are generally efficient but exhibit the classic favorite-longshot bias in the low-probability tail. Offering a wider range of thresholds around the expected unemployment rate could reduce this bias. For forecasters, our results suggest that prediction market prices should be interpreted with caution for low-probability unemployment events, as the market tends to overprice these events. Prices in the 0.30 to 1.00 range, however, can be interpreted with confidence.

Future research should examine whether similar patterns emerge in other prediction markets for economic indicators, such as GDP or inflation. Additionally, research that examines the dynamics of the favorite-longshot bias over time could provide insight into whether the bias is stable or varies with market conditions. Finally, research that tests the profitability of trading strategies based on the favorite-longshot bias after accounting for transaction costs would provide practical guidance for market participants.

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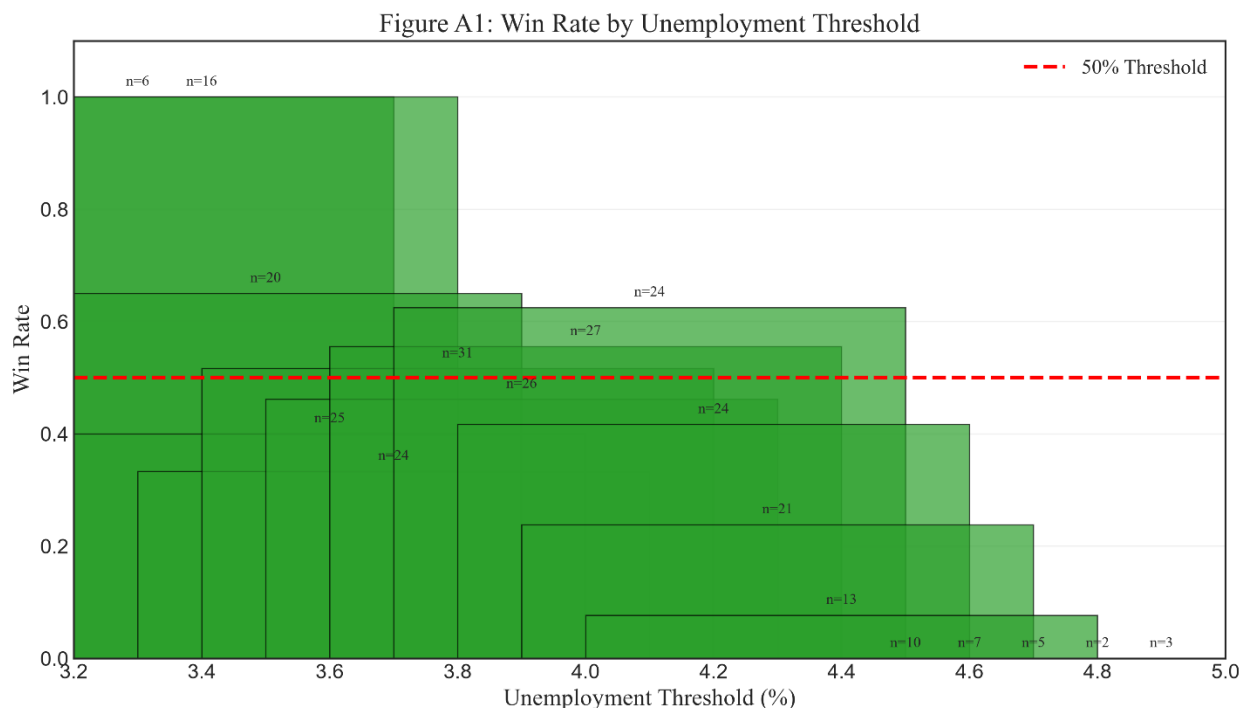
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Appendix A: Supplementary Figures

This appendix contains a supplementary figure that supports the main analysis presented in the paper. This figure provides additional visual evidence for the threshold effects and price dynamics discussed in Section 4.

Figure A1: Win Rate by Unemployment Threshold



Note: The bar chart displays the win rate for each unemployment threshold level. The horizontal dashed line at 0.50 represents the 50% threshold. Sample sizes (n) are shown above each bar.

The distribution of final prices reveals a relatively balanced pattern with some concentration at the extreme. The figure shows a negative relationship between threshold and win rate, indicating that the market accurately priced the declining probability of higher unemployment outcomes. Thresholds below 3.8% have win rates at or above 50%, reflecting the market's correct assessment that unemployment would exceed these levels. Thresholds between 3.9% and 4.1% show mixed outcomes, with win rates ranging from 0% to 100%, representing the region of greatest uncertainty. Thresholds above 4.2% have win rates at or near 0%, indicating that the market correctly identified these outcomes as impossible. This pattern is consistent with the calibration analysis in Section 4.2, where the market showed systematic overpricing in the low-probability range.